Q-1- What are the different protocols you observe at the following layers of the protocol stack?

Ans→

Application Layer	Transport layer	Network layer
MDNS	UDP	ARP
HTTP	TCP	ICMPV6
SSDP		IGMV3
DHCP		
DNS		

Q-2- What is the total amount of data being received for the following two cases?

A) When you access :- http://neverssl.com

B) When you access :- https://www.cornell.edu ip.addr==10.85.17.19

No.	Time	Source	Destination	Protocol	Length	Info
1312	15.465927095	10.85.17.19	255.255.255.255	UDP	867	47209 → 29810 Len=825
3739	45.569314312	10.85.17.19	255.255.255.255	UDP	867	47209 → 29810 Len=825
413	5.429738891	10.85.17.19	255.255.255.255	UDP	868	47209 → 29810 Len=826
2153	25.500036425	10.85.17.19	255.255.255.255	UDP	868	47209 → 29810 Len=826
2912	35.534306722	10.85.17.19	255.255.255.255	UDP	868	47209 → 29810 Len=826
5496	65.639501682	10.85.17.19	255.255.255.255	UDP	868	47209 → 29810 Len=826
6326	75.674323473	10.85.17.19	255.255.255.255	UDP	868	47209 → 29810 Len=826
4680	55.604466191	10.85.17.19	255.255.255.255	UDP	869	47209 → 29810 Len=827

Q-3 Answer the following when you access the site http://neverssl.com

A) How many HTTPGET requests do you observe? List down the GET requests.

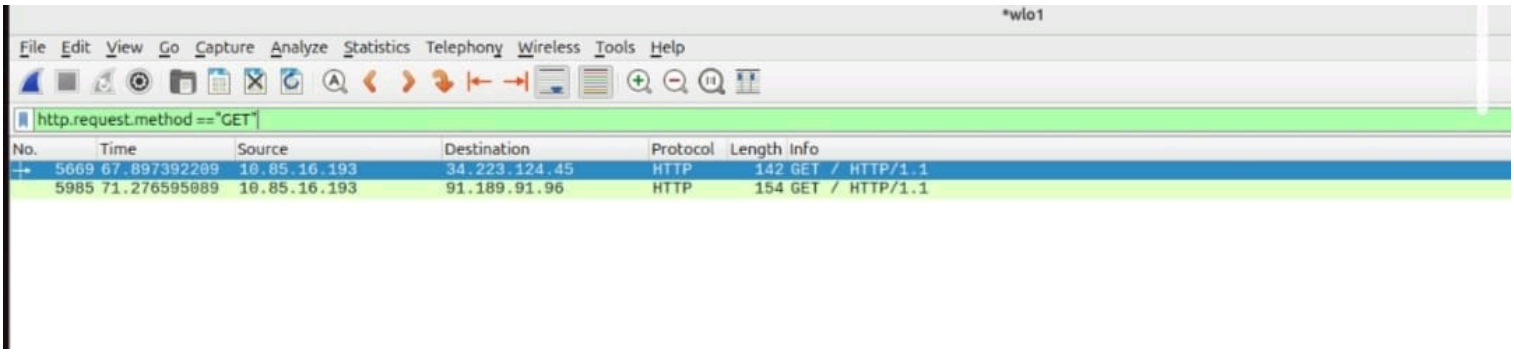
Ans→ Two HTTP GET requests observed.

GET requests :-

Time	Source	Destination	protocol	Length	info
5669 67.897392209	10.85.16.193	34.223.124.45	HTTP	142	GET/HTTP/1.
5985 71.276595089	10.85.16.193	91.189.91.96	HTTP	154	GET/HTTP

B) what version of HTTP the server is running?

GET/Http/1.1


*wlo1
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
http.request.method =="GET"
No. Time Source Destination Protocol Length Info
5669 67.897392209 10.85.16.193 34.223.124.45 HTTP 142 GET / HTTP/1.1
5985 71.276595089 10.85.16.193 91.189.91.96 HTTP 154 GET / HTTP/1.1

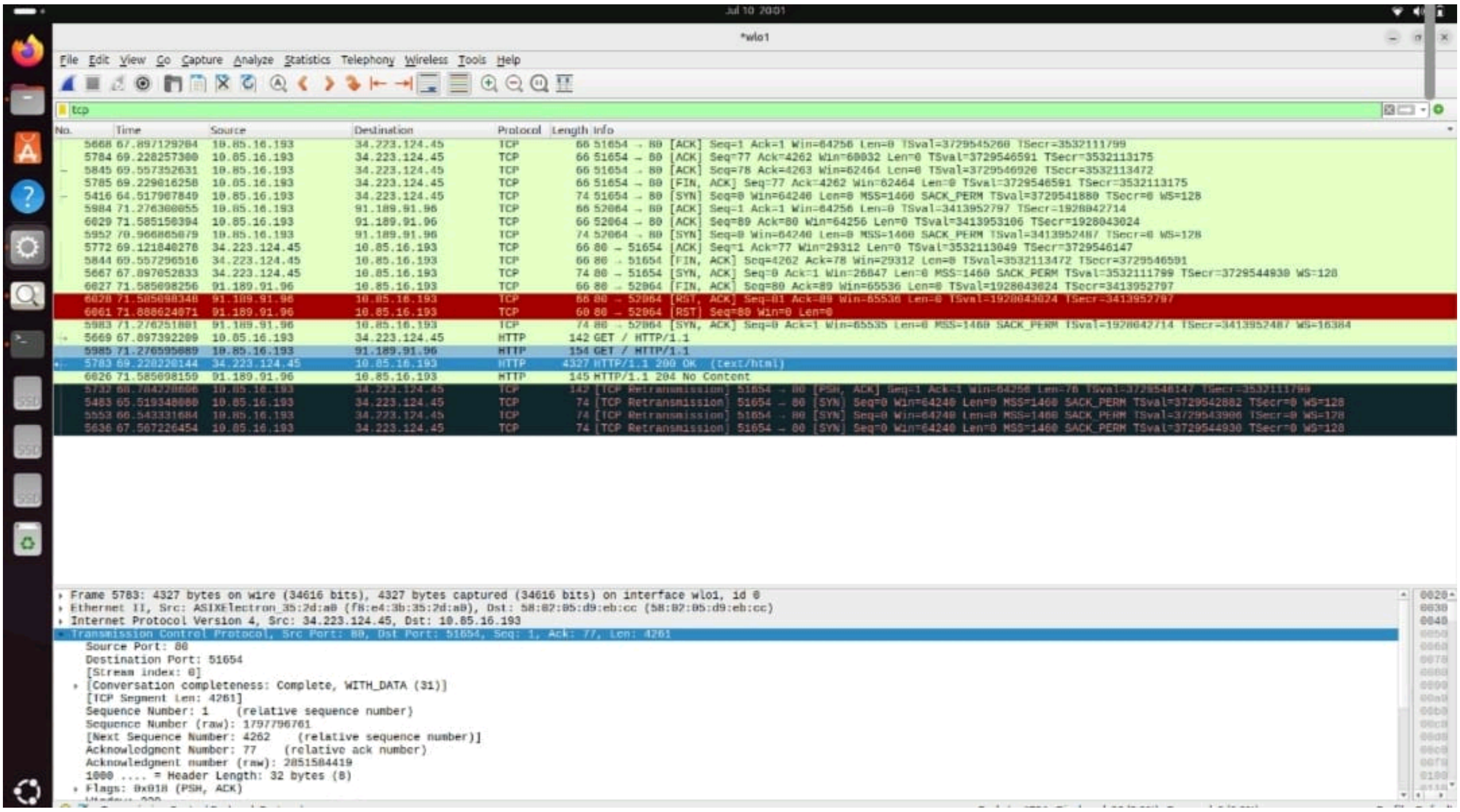
c) For each of the Http Get request find out:

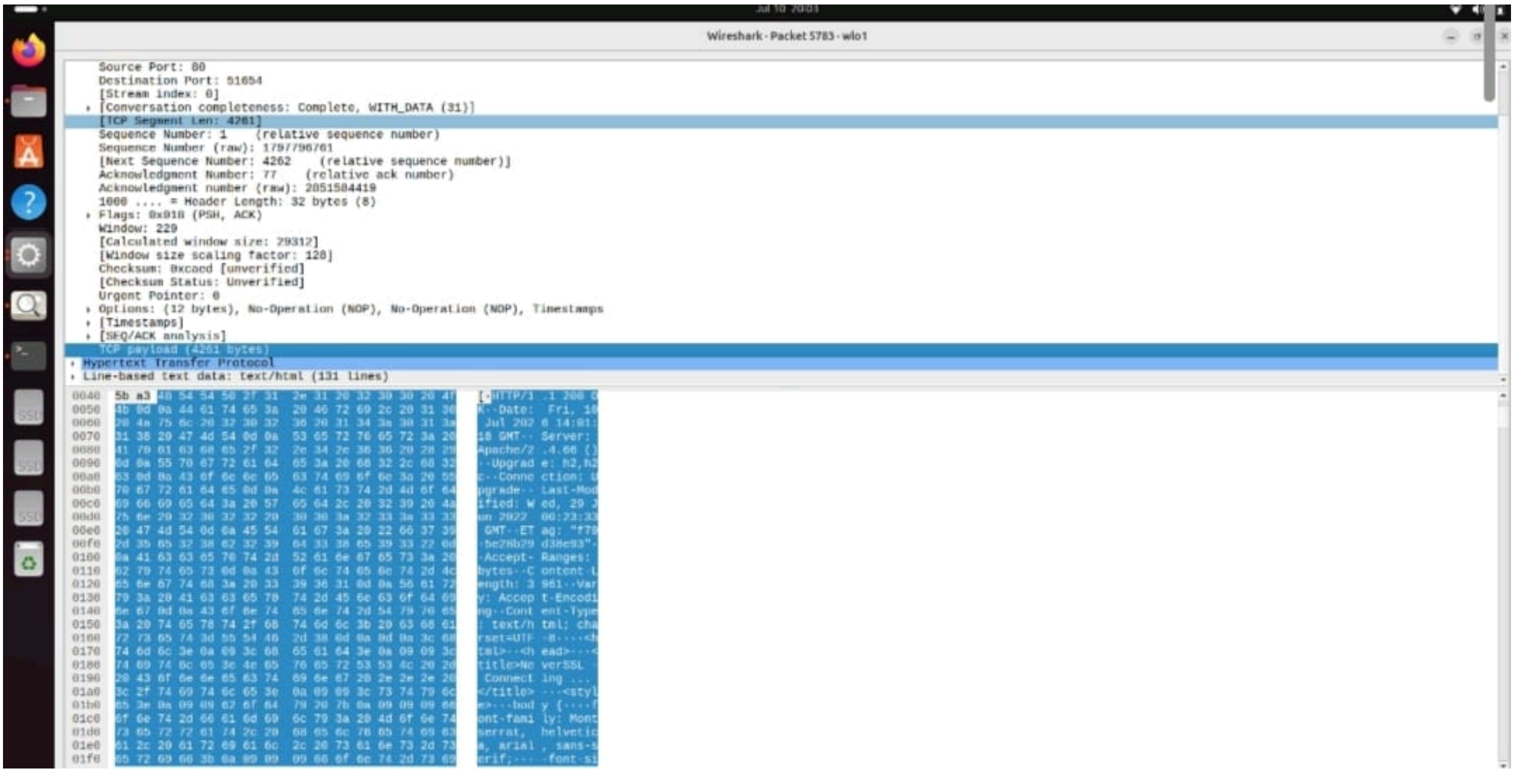
i) The total number of TCP segments being recieved

Ans→ one TCP segment recieved

ii) The total amount of data being received in the corresponding HTTP response message.

Ans→ 4261 bytes.


*wlo1
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
5668 67.897129204 10.85.16.193 34.223.124.45 TCP 66
5784 69.228257300 10.85.16.193 34.223.124.45 TCP 66
5845 69.557352631 10.85.16.193 34.223.124.45 TCP 66
5785 69.229016258 10.85.16.193 34.223.124.45 TCP 66
5416 64.517967849 10.85.16.193 34.223.124.45 TCP 74
5984 71.276300855 10.85.16.193 91.189.91.96 TCP 66
6029 71.585150394 10.85.16.193 91.189.91.96 TCP 66
5952 70.966865079 10.85.16.193 91.189.91.96 TCP 74
5772 69.121840278 34.223.124.45 10.85.16.193 TCP 66
5844 69.557296516 34.223.124.45 10.85.16.193 TCP 66
5667 67.897052833 34.223.124.45 10.85.16.193 TCP 74
6027 71.585098256 91.189.91.96 10.85.16.193 TCP 66
6061 71.888624071 91.189.91.96 10.85.16.193 TCP 60 80 → 52064 [RST] Seq=80 Win=0 Len=0
5669 67.897392209 10.85.16.193 34.223.124.45 HTTP 142 GET / HTTP/1.1
5985 71.276595889 10.85.16.193 91.189.91.96 HTTP 154 GET / HTTP/1.1
5783 69.228220144 34.223.124.45 10.85.16.193 HTTP 4327 HTTP/1.1 200 OK (text/html)
6026 71.585098159 91.189.91.96 10.85.16.193 HTTP 145 HTTP/1.1 204 No Content
5483 65.519348080 10.85.16.193 34.223.124.45 TCP 74 [TCP Retransmission] 51654 → 80 [SYN]
5636 67.567226454 10.85.16.193 34.223.124.45 TCP 74 [TCP Retransmission] 51654 → 80 [SYN]
Frame 5783: 4327 bytes on wire (34616 bits), 4327 bytes captured (34616 bits) on interface wlo1, id 0
Ethernet II, Src: ASIXElectron_35:2d:a0 (f8:e4:3b:35:2d:a0), Dst: 58:02:05:d9:eb:cc (58:02:05:d9:eb:cc)
Internet Protocol Version 4, Src: 34.223.124.45, Dst: 10.85.16.193
Transmission Control Protocol, Src Port: 80, Dst Port: 51654, Seq: 1, Ack: 77, Len: 4261
Source Port: 80
Destination Port: 51654
[Stream index: 0]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 4261]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1797796761
[Next Sequence Number: 4262 (relative sequence number)]
Acknowledgment Number: 77 (relative ack number)
Acknowledgment number (raw): 2851584419
1000 .... = Header Length: 32 bytes (8)
Flags: 0x018 (PSH, ACK)


Wireshark · Packet 5783 · wlo1
Source Port: 80
Destination Port: 51654
[Stream index: 0]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 4261]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1797796761
[Next Sequence Number: 4262 (relative sequence number)]
Acknowledgment Number: 77 (relative ack number)
Acknowledgment number (raw): 2851584419
1000 .... = Header Length: 32 bytes (8)
Flags: 0x018 (PSH, ACK)
Window: 229
[Calculated window size: 29312]
[Window size scaling factor: 128]
Checksum: 0xcoed [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps
[Timestamps]
[SEQ/ACK analysis]
TCP payload (4261 bytes)
Hypertext Transfer Protocol
Line-based text data: text/html (131 lines)

1281 bytes.

D) Identify TCP three way handshake used to establish the connection with the http://neverssl.com. Specify SYN, SYN-ACK, ACK packets and write the sequence number for each of the three accesses.

Ans → • SYN packet:-

Source IP:- 10·85·17·18 (client)

Destination IP:- 34·223·124·45 (Server)

protocol info:- 57022 → 80 [SYN]

• SYN-ACK packet:-

Source IP:- 34·223·124·45 (Server)

Destination IP:- 10·85·17·18 (Client)

protocol info:- 80 → 57022 [SYN, ACK]

Sequence Number:- 0

(ACK = 1)

• ACK packet:-

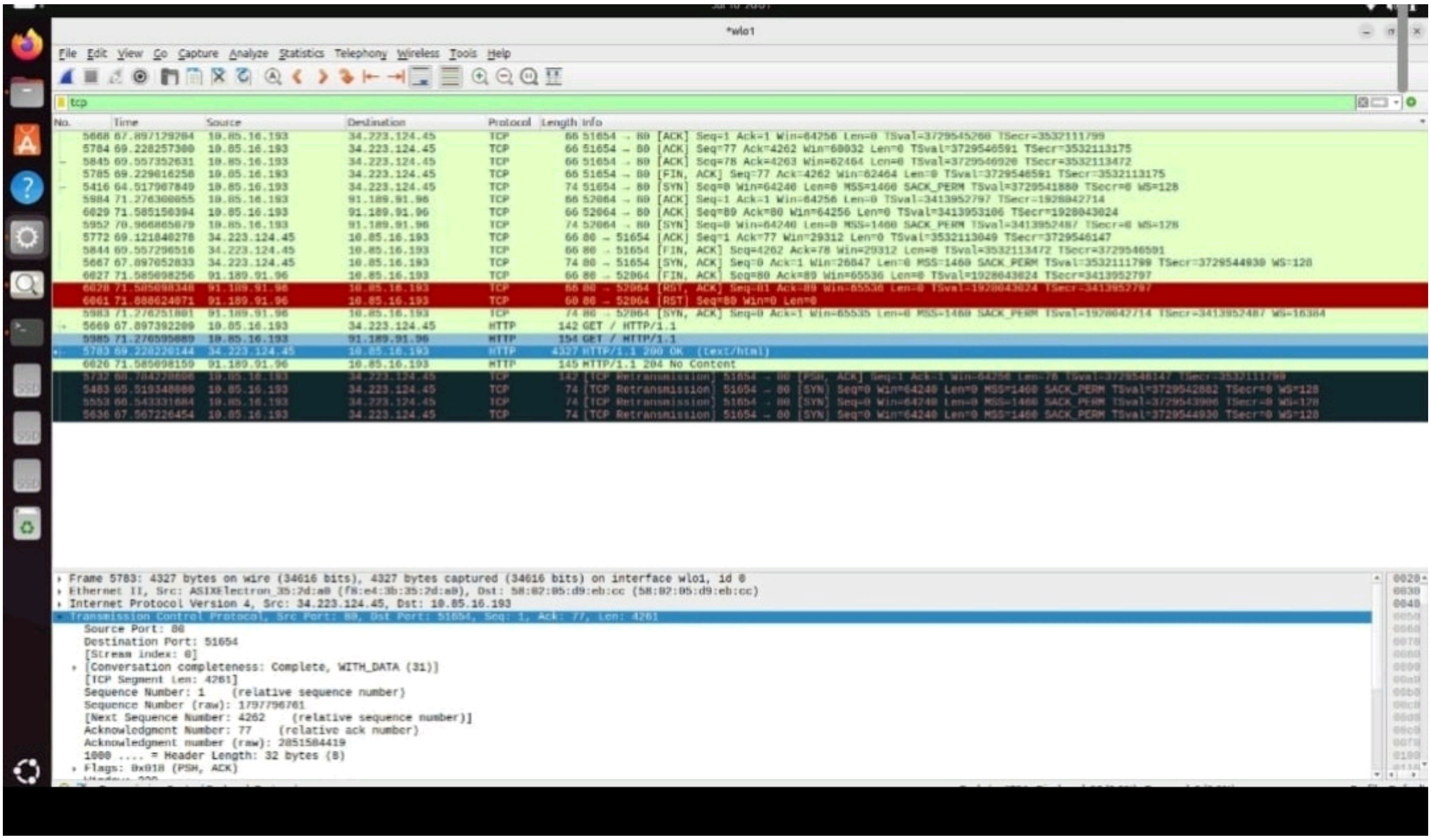
Source IP:- 10·85·17·18 (Client)

Destination IP:- 34·223·124·45 (server)

protocol Info 57022 → 80 [ACK]

Sequence Number:- 1

(ACK = 1)


*wlo1
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
Frame 5783: 4327 bytes on wire (34616 bits), 4327 bytes captured (34616 bits) on interface wlo1, id 0
Ethernet II, Src: ASIXElectron_35:2d:a0 (f8:e4:3b:35:2d:a0), Dst: 58:02:05:d9:eb:cc (58:02:05:d9:eb:cc)
Internet Protocol Version 4, Src: 34.223.124.45, Dst: 10.85.16.193
Transmission Control Protocol, Src Port: 80, Dst Port: 51654, Seq: 1, Ack: 77, Len: 4261
Source Port: 80
Destination Port: 51654
[Stream index: 0]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 4261]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 1797796761
[Next Sequence Number: 4262 (relative sequence number)]
Acknowledgment Number: 77 (relative ack number)
Acknowledgment number (raw): 2051584419
1000 .... = Header Length: 32 bytes (8)
Flags: 0x018 (PSH, ACK)